Crossing the Assembly Theory Biological Threshold: A Computational Proof via MEA Architecture

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Abstract

Assembly Theory (AT) posits that biological life can be objectively distinguished from abiotic matter by quantifying the recursive steps required to assemble a complex molecule, defining a threshold (Assembly Index > 15) unique to living systems. In this paper, we present a computational proof crossing this threshold using the MEA (Multiverse Evolutionary Architecture) Engine. By integrating quantum-inspired thermodynamics, Numba Just-In-Time (JIT) compilation, and evolutionary polymer populations, our simulation achieved an Assembly Index of 99 within 5.07 seconds, generating 17 highly complex biological structures. Deterministic reproducibility is strictly maintained, and data immutability is guaranteed via NIST FIPS 180-4 compliant SHA-256 cryptographic signatures. This demonstrates that informational selection in abiotic environments can dynamically breach the threshold of life *in silico*.

Keywords: Assembly Theory, MEA Core, Quantum Thermodynamics, Artificial Life, Cryptographic Auditing, Reproducibility.

1 Introduction

The transition from inert matter to living systems is one of the most profound open questions in science. Recently, Assembly Theory (Cronin, Walker *et al.*) introduced a measurable framework to distinguish biological signatures from random noise. The core metric, the Assembly Index (A_I), represents the shortest pathway to construct a molecule from basic building blocks. Empirical evidence suggests that objects with an $A_I > 15$ cannot occur in significant abundance without a directed, life-like informational engine.

This paper proposes an algorithmic approach using the MEA (Multiverse Evolutionary Architecture) Engine to computationally simulate the crossing of the abiotic-biotic threshold, demonstrating how harmonic dissipative adaptation drives molecular complexity.

2 Methodology

2.1 The MEA QuantumBrain Architecture

The core simulation was executed on the MEA v14.5 engine, specifically utilizing the `QuantumBrain` and `QuantumFractalMemory` (Q-RAG) modules. The environment consisted of polymer chains of size $N = 64$ simulated over 1,500 continuous topological iterations.

Mutation rates were harmonically modulated based on the thermodynamic entropy state of the `QuantumBrain`, keeping the mutation rate between 1.5% and 6%. The intensive computations were optimized using Numba Just-In-Time (JIT) fastmath directives, achieving hardware-level C-speed execution in Python.

2.2 Strict Reproducibility and Cryptographic Auditing (NIST Standard)

In computational sciences, reproducibility is the gold standard. To ensure that the results are completely deterministic and verifiable by third parties, the pseudorandom number generator was fixed using `np.random.seed(2026)`.

Furthermore, the system employs an automated cryptographic ledger. Upon crossing the biological threshold, the resulting metrics are serialized and hashed using the SHA-256 algorithm (NIST FIPS 180-4 compliant). The resulting hash and the system's coherence state (Θ) are securely stored in the local `blueprints_vault.json` ledger.

3 Results

The simulation was executed in a standard computing environment. The execution time to simulate 1,500 evolutionary generations over the recursive assembly space was strictly **5.07 seconds**.

The final highest Assembly Index recorded was $A_I = 99$, substantially surpassing the established biological threshold of 15. Furthermore, the abundance parameter yielded 17 active deep-assembly structures, confirming that the system did not rely on a random statistical anomaly, but rather an active thermodynamic selection process.

Table 1: Simulation Outputs and Assembly Metrics

Metric	**Value**
Execution Time	5.07 s
Assembly Index (A_I)	99
Complex Structures Abundance	17
Quantum Coherence (Θ)	1.5708 rad
Q-RAG Node Location	`[qnode_0182]`
Cryptographic Hash (SHA-256)	`fa615c6b...`
Biotic Status	Confirmed

4 Discussion

The achievement of $A_I = 99$ implies that the MEA engine successfully replicates the physical constraints and evolutionary drive required to cross the origin-of-life barrier *in silico*. The explicit cryptographic signature (`fa615c6b2e64c5a0d202a3e7e73e8ba5...`) seals the state of the experiment, paving the way for trustless scientific benchmarking. Because the system was initialized with a fixed state, any independent researcher running the MEA Core module will arrive at the exact same A_I and SHA-256 signature, definitively proving the deterministic nature of this complexity generation.

5 Conclusion

We have demonstrated a computational bridging of the Assembly Theory threshold using the MEA thermodynamic engine. By achieving a sustained biological signature rapidly, audibly, and reproducibly, this framework serves as a robust new paradigm for Origins of Life simulation and biomimetic algorithms.

Code Availability

The executable code (`assembly_theory_life_mea.py`), cryptographic logs, and video proofs of execution are fully available at the author's public GitHub repository to allow broad peer-review and execution.